ISTA 130 Mutable-Immutable Worksheet

Write a function called `upper_list` that takes a list of strings as its sole argument and converts them all to uppercase. The function should alter the original list and return `None`.

Write a function called `upper_list2` that takes a list of strings as its sole argument. It returns a new list containing all of the strings from the original list in uppercase. The function should NOT alter the original list.

Write a function called `upper_values` that takes a dictionary as its sole argument. This dictionary maps keys, which may be any immutable object, to values, which are lists of strings. The function converts all of the strings in all of the lists to uppercase (ie the dictionary values). It should alter the original lists. What should the function return? You may call the previous functions in your code. You should start by drawing a picture of the dictionary so its structure is clear in your mind.

Write a function called `upper_values2` that takes a dictionary as its sole argument. This dictionary maps keys, which may be any immutable object, to values, which are lists of strings. It returns a new dictionary. The new dictionary is identical to the original dictionary except that all of the strings in all of the lists are in uppercase (ie the dictionary values). The function should **NOT** alter the original dictionary. You may call the previous functions in your code.